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Determinants of Environmental Law Violation Fines Against Petroleum Refineries: Race, Ethnicity, Income, and Aggregation Effects

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In this work we determine whether monetary penalties assessed against petroleum refineries for violation of the Clean Air Act, Clean Water Act, and/or the Resource Conservation and/or Recovery Act differ depending upon the racial, ethnic, and income characteristics of communities surrounding the penalized refinery. Our sample consists of all monetary penalties assessed between April 2001 and April 2003 ($n = 206$). We find mixed evidence of inequality. The racial, ethnic, and income characteristics of census tracts surrounding the penalized refineries are not related to penalty amounts. However, refineries situated within the boundaries of Hispanic and low-income ZIP codes tend receive smaller penalties than refineries located in non-Hispanic and more affluent ZIP codes. These disparities do not appear to be result of factors such as previous enforcement and compliance history, case characteristics, company and facility characteristics, or the political/economic climate.

Keywords African American, aggregation errors, compliance and enforcement, environmental crime, environmental decision making, environmental justice, Hispanic, income, penalties, petroleum refineries, U.S. Environmental Protection Agency

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Environmental justice is defined as the “fair treatment of all races, cultures, incomes, and educational levels with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies” (Rhodes 2003, 19). Empirical studies of environmental justice, however, tend to assess the spatial relationships between race, ethnicity, and the location of hazardous waste facilities or the release of toxic chemicals (Allen 2001; Anderson, Anderton, and Oakes 1994; Pollock and Vittas 1995). This article examines whether race and/or economic characteristics of an area affect the monetary penalties assessed against petroleum refineries for violations of federal environmental laws. The examination of penalty amounts is an important contribution to the environmental justice literature for two reasons. First, monetary penalties are believed to deter environmental non-compliance (Mintz 1995, 10–11). Thus, if penalties meted out for violation of environmental law vary by community characteristics, corporations may be more willing to violate laws in “less protected” areas. Second, if monetary penalties for similar violations are lower in poor and minority communities than in White and affluent communities, then environmental law enforcement agencies are not meeting their commitment to environmental justice (Bullard 1994).

Prior Research

Three studies specifically address the association between race, ethnicity, income and penalties against entities that violate environmental regulations. Lavelle and Coyle (1992) examined the U.S. EPA’s Civil Enforcement DOCKET database to investigate whether potential race and class biases existed in the distribution of penalties for violations of the Clean Air Act (CAA), Clean Water Act (CWA), and Resource, Conservation and Recovery Act (RCRA) for 1985–1991. They found that the average fine for violating federal environmental laws was higher in White (\$153,067) compared to minority ZIP codes (\$105,028).

Ringquist (1998) and Atlas (2001) reanalyzed the Civil Enforcement DOCKET database in an effort to replicate Lavelle and Coyle’s findings. Ringquist, who studied ZIP code demographics, was able to replicate Lavelle and Coyle’s bivariate findings. However, using ordinary least-squares regression and additional control variables, Ringquist did not find evidence of inequality. Using census-tract-level variables, Atlas (2001) discovered that minority census-tract areas actually receive *higher* fines than White areas (\$133,808 vs. \$113,791), and that low-income census-tract areas receive *higher* fines than high-income areas (\$162,804 vs. \$82,816). According to Atlas, several issues need to be addressed in future studies of environmental justice and penalties assessed for environmental violations.

First, information contained in DOCKET is insufficient for examining environmental justice issues. One reason is that DOCKET does not account for state penalties, which make up a large number of enforcement actions. Second, DOCKET data are missing several control variables that may impact fine amounts. For instance, Atlas (2001) argues that researchers should control for past noncompliance and prior enforcement history. In addition, regional variations in enforcement policy should also be controlled. For example, the General Accounting Office (U.S. GAO 2000, 5) noted large differences in the “size of penalties assessed and the criteria used in determining penalties assessed” by U.S. EPA region, and that “enforcement responses in certain regions are weaker than they are in others” (U.S. GAO 2000, 8). In addition, community race and ethnic characteristics are likely to vary by U.S. EPA region. Thus, failure to control for regional differences may

produce misleading results with respect to environmental justice analyses of penalty variations.

Petroleum Refineries

Why focus on the petroleum industry? Petroleum is not only a significant portion of the U.S. economy; it also produces a significant amount of pollution. According to the U.S. Department of Commerce, Bureau of the Census (1999), petroleum refineries produce nearly \$160 billion in shipments or about 4% of all manufacturing shipment values. Refineries are primarily concentrated in three states—Texas, Louisiana, and California—which produced 56% of shipment values.

Petroleum refineries also produce a great deal of pollution. According to the most recent Toxic Release Inventory (TRI), 67% of these pollutants are released into the air while 24% are released to water (U.S. EPA, 2003b). Petroleum refineries produce significantly more air pollution and slightly less water pollution than the average industrial facility. In 2001 petroleum refineries released approximately 78 million pounds of regulated chemicals into the environment. This represents nearly 11.5% of all chemical releases reported to the TRI. In addition, petroleum refineries report almost three times more chemical releases than the average TRI reporting facility. Further, 26% of all reported TRI releases are volatile organic compounds (VOCs) and are a significant public health concern. For instance, benzene, a carcinogenic VOC, is commonly released during petroleum refining (U.S. EPA, Office of Compliance 1995). Studies suggest increased mortality rates and cancer risk for petroleum refinery workers and residents living near petroleum refineries as a result of benzene releases (Hurtig and San Sebastian 2002; Ireland et al. 1997; Rinsky et al. 1987; Tsai and Wendt 2001). Finally, petroleum refineries are the largest single source of greenhouse gas emissions.

In terms of environmental behavior, the petroleum refinery industry is one of the most regulated and most persistent violators of federal environmental laws. Nearly all facilities are inspected annually and approximately 25% of all inspections result in a state or federal enforcement action (U.S. EPA, Office of Compliance 1995, 110). Between 1990 and 1995, the percentage of enforcement actions against petroleum refineries that resulted from inspections was greater for the petroleum industry than for any other industry except electronics. During that time period a total of 797 enforcement actions occurred at approximately 71% of petroleum refineries.

Data and Methods

This study examines whether environmental law penalties assessed against petroleum refineries differ depending upon the racial, ethnicity, and income characteristics of communities surrounding the penalized refinery. To construct the sample for this research, we identified all 153 operable petroleum refineries during 2001 using the U.S. Department of Energy, Energy Information Administration (2002) *Petroleum Supply Annual 2001* Vol. 1. Next, compliance and enforcement information for all refineries was obtained using the U.S. EPA's (2003) Enforcement and Compliance System Online (ECHO). That database allowed us to obtain a list of all the monetary penalties assessed against petroleum refineries between April 2001 and April 2003 for violations of the CAA, CWA, and RCRA. ECHO was also employed to examine the compliance and enforcement history of each refinery.

A total of 294 state and federal enforcement actions resulted in penalties assessed against 99 of the 153 facilities on our list. Thirty-four refineries appeared to have multiple fines for a single incident. These cases were often reported as “combined” by the U.S. EPA, and penalties were thus summed and reported as a single enforcement action that resulted in a single penalty. In all combined cases only one of the cases had a nonzero dollar penalty amount. A few refineries had penalties that resulted from a violation for the same law on the same day. These penalty amounts were also summed and treated as one enforcement action that resulted in a single penalty. This procedure produced a sample of 206 enforcement actions.

The data used to create the dependent variable (assessed penalty amount), the independent variables (e.g., racial and economic characteristics of areas surrounding the petroleum refineries), and control variables were obtained from Dun and Bradstreet (2003) *Million Dollar Database*; the U.S. Department of Commerce, Bureau of Census (2002) *2000 Census of Population and Housing, Summary File 3*; Environmental Defense (1999) *Pollution Prevention Performance Ranking for Oil Refining Facilities*; and the U.S. EPA (2003a) *ECHO* database.

Dependent Variable

The dependent variable in this study is fine amount assessed against petroleum refineries that violated CAA, CWA, or RCRA between April 2001 and April 2003. It is important to note that not all enforcement actions result in monetary penalties. During this period, the U.S. EPA and state regulatory agencies assessed fines that ranged between \$0 and \$9,999,999. Nearly 50% of all fines were for amounts less than \$15,000.

Community Demographics

Theoretically, penalties assessed to petroleum refineries should be based on legally relevant variables and not on extralegal factors. To determine if this is true, measures of extralegal variables must be obtained. To do so, latitude and longitude for each facility (as listed in ECHO) were imported into Arcview 3.2 to determine the 2000 census-tract location of each petroleum refinery. Once refineries were mapped, demographic data were obtained from the *2000 Census of Population and Housing* for each refinery census tract (U.S. Department of Commerce, Bureau of Census 2002). Consistent with previous environmental justice research, three census-tract demographic measures were included as indicators of extralegal factors that may influence penalty amounts: (1) the proportion of African American residents; (2) the proportion of Hispanic residents; and (3) median tract household income (in thousands of dollars).

There has been considerable debate concerning the most appropriate unit of analysis in environmental justice studies (Liu 2001). For instance, some researchers prefer census tracts because they more closely approximate community characteristics than larger units (Streetsky and Hogan 1998), while others argue that ZIP codes are more appropriate units from which to draw comparisons (Goldman and Fitton 1994). With respect to the distribution of environmental penalties, Lavelle and Coyle (1992) and Ringquist (1998) studied ZIP-code demographics while Atlas (2001) studied groups of census tracts.

In order to determine whether there is any consistency across units of analysis, we examine the relationship between community racial, ethnic, and

economic characteristics and petroleum refinery penalties at both the census-tract and ZIP-code levels.

Control Variables

Compliance and Enforcement History

Traditional criminal justice research indicates that past criminal and deviant behavior is an important predictor of sentencing (Kramer and Steffensmeir 1993). We measure compliance history using two variables: (1) previous history of significant noncompliance (SNC) or high priority violations (HPV) and (2) past state and U.S. EPA enforcement actions at the refinery.

Designation as an SNC or HPV identifies whether a refinery has a recent history of serious violations with the CAA (HPV) or CWA and RCRA (SNC). To be designated as SNC or HPV, one of several conditions must be met: A refinery failed to submit a major report; far exceeded the permitted discharges; caused actual exposure (or a substantial likelihood of actual exposure) to hazardous waste; or is a reckless violator that deviates substantially from the terms of an arrangement (i.e., permit, order, or agreement) with the U.S. EPA. Nearly 52.6% of all petroleum refineries penalized for environmental violations were in significant noncompliance for violations of the CAA, CWA, or RCRA between 1999 and 2001. SNC and HPV were measured for the 2 years prior (1999–2001) to the data employed in the present analysis. SNC/HPV facilities were coded as 1, while those without compliance or violation histories were coded 0.

The number of enforcement actions initiated by the U.S. EPA and state governments against each facility between 1999 and 2001 was also employed as a measure of past violations. Theoretically, enforcement actions are likely to be initiated when violations are discovered, and may also indicate a pattern of recidivism. The U.S. EPA took previous enforcement actions at approximately 65.3% of the 99 petroleum refineries in our sample.

Case Characteristics

Several case characteristics relevant to setting penalty amounts are available in ECHO data. First, the type of law violated may have an important impact on the assessed penalty because average fines are likely to vary across statutes (Atlas, 2001; Ringquist 1998). Statute specific effects are controlled for by including dummy variables for the CAA, CWA, and RCRA.

Multimedia cases involve either multistatute or multifacility violations. Cases of multistatute violations are likely to invoke a larger penalty because they involve a number of different environmental threats. Multifacility violations are also likely to be treated more seriously because the extent of the threat to humans and the environment is typically greater than single facility violations. We create a dichotomous variable and code it 1 if it is multimedia and 0 if it is not multimedia. Not all cases that occurred at multiple facilities were treated as multimedia cases. Therefore, we also control for the number of penalty locations by including a variable that measures the total number of facilities covered by that penalty.

Evidence indicates that case severity may influence whether a state agency or the U.S. EPA will take the lead in an enforcement action (Mintz 1995). Case severity, in turn, may be related to the amount that a violator is fined. For example, state and

local prosecutors have accused the U.S. EPA of ignoring minor environmental violations—where penalties tend to be lower—and only taking the lead on the most serious cases (Mintz 1995). Thus, we include a dichotomous variable indicating whether the state (coded 0) or the U.S. EPA (coded 1) had the lead in the enforcement action.

Finally, we control for whether the case is resolved through judicial action (coded 1) or through a less formal order or negotiation (coded 0) to assess whether cases resolved through judicial means have larger fines assessed than cases resolved through nonjudicial means.

Company and Facility Characteristics

Company characteristics may impact the penalty assessed against petroleum refineries. For example, corporate crime research suggests that bigger companies may be able to better resist regulatory enforcement (Clifford 1998). There is less agreement, however, as to whether large corporations are more likely to violate environmental laws (Baron 1984). Furthermore, because the U.S. EPA penalty system is designed to achieve compliance and deterrence, fines must be sufficiently large to affect decision-making. To do so, fines must be proportionate to economic rewards such as profits and income. We measure company size using two variables obtained from the Dun and Bradstreet *Million Dollar Database*: (1) total annual company sales (in billions of dollars) and (2) the total number of company employees (in thousands of employees). That database provides detailed information about a variety of public and private businesses and is considered to be a reliable source of financial information.

We also determine whether the penalized company is a subsidiary of another company. Case studies suggest that subsidiaries are able to draw upon the resources of parent companies but at the same time are able to shield parent companies from some liability (Prechel, Boies, and Wood 1999). Thus, companies that have branches that regularly violate environmental laws may see some benefit from obtaining subsidiary status. Subsidiaries are coded as 1 and nonsubsidiaries are coded as 0.

Less pollution efficient companies may be more likely to violate the law and draw more substantial penalties as a result of their mode of operation. To determine if less efficient companies are assessed higher penalties than more efficient companies we obtained the *Pollution Prevention Performance Ranking for Oil Refining Facilities* from Environmental Defense (1999). This list ranked a company's refineries into three groups according to the amount of toxic wastes and criteria pollutants generated for each barrel of crude oil processed in 1997. Facilities that had the worst pollution prevention performance were assigned the score of 3; those that had average pollution prevention performance were assigned the score of 2; and, those that had the best pollution prevention performance were assigned the score of 1.

Political and Economic Climate

The political climate of the state may be important in assessing penalties against petroleum refineries. Some researchers argue that there is little difference between mainstream political parties since both democrats and republicans actively seek out industrial development (Cobb 1993). Other researchers, however, assert that political climate of the state may be important (Mintz 1995). We measure state environmental preferences by determining the political party affiliation of the state governor at the time of the fine. Likewise, the petroleum refinery industry is powerful and may

be able to use its influence to mitigate the effectiveness of enforcement actions (Ringquist 1998, 1158). Following McKinney (1998, 386), we hypothesized that “Democratic governors are likely to be more liberal on environmental matters than Republicans,” and less susceptible to petroleum refinery lobbying efforts. These governors might be able to influence policy. Areas in states with Republican governors were coded 0, while areas in Democratic states were coded 1.

Variation in industry strength for each state in which a refinery is located was measured by determining the value added by the petroleum refinery industry (Standard Industrial Classification code 2911) as a percentage of the gross state product during 1997. This data was gathered from the U.S. Department of Commerce, Bureau of the Census (1999) *1997 Economic Census Manufacturing Industry Series: Petroleum Refineries* and the U.S. Department of Commerce, Bureau of Economic Analysis (2003) web site.

A recent report released by the U.S. General Accounting Office (2000) indicates that there is considerable enforcement inconsistency among the 10 U.S. EPA regional offices. Moreover, even though the U.S. EPA influences environmental enforcement through its headquarters by issuing various policies and memorandums to guide enforcement policy, the U.S. GAO found considerable differences in philosophical approaches to enforcement between regional offices. Differences in philosophies and other factors such as budgets result in variations in enforcement practices. For instance, according to the U.S. GAO, regional variations in CAA-related facility inspections ranged from a low of 27% to a high of 74% (U.S. GAO 2000, 7). In order to control for regional variations in enforcement patterns as well as variations in racial, ethnic, and economic composition, we created a dummy variable to represent the U.S. EPA region where a petroleum refinery is located (Region 1—CT, MA, ME, RI, NH, and VT; Region 2—NJ, NY, PR, and VI; Region 3—Washington DC, DE, MA, PA, VI, and WV; Region 4—AL, FL, GA, KY, MS, NC, SC, and TN; Region 5—IL, IN, MI, MN, OH, and WI; Region 6—AR, LA, NM, OK, and TX; Region 7—IA, KS, MO, and NE; Region 8—CO, MT, ND, SD, UT, and WY; Region 9—AZ, CA, HI, and NV; Region 10—AK, ID, OR, and WA). No penalties were assessed against refineries in Region 1, so Region 2 was employed as the omitted comparison category.

Analysis and Results

The first step in our analysis compared the mean and median penalties against petroleum refineries according to three groupings for both census tracts and ZIP codes: (1) African American versus White census tracts; (2) Hispanic versus non-Hispanic tracts; and (3) high- versus low-income tracts. To be consistent with previous research on U.S. EPA penalties we examine mean and median comparisons between cases that fall below the lower quartile (i.e., bottom 25% of the cases in the distribution) and cases that fall above the upper quartile (i.e., top 25% of the cases in the distribution) for median income, percent African American, and percent Hispanic (see Lavelle and Coyle 1992; Ringquist 1998; and Atlas 2001). The results of that analysis are displayed in Table 1. To determine if observed differences in penalty amounts between the two groups were statistically meaningful, difference of means (*t*-tests for independent samples) and medians (Mann–Whitney *U*) tests were employed.

As Table 1 indicates, the bivariate results are highly inconsistent across different units of aggregation. Specifically, for *census tracts*, mean and median penalties

TABLE 1 Comparison of Mean and Median Penalties Assessed Against Petroleum Refineries Within Quartiles for Proportion African American, Proportion Hispanic, and Median Household Income ($n = 104$)

			Mean difference (t -score; p value)	Median difference (z -score; p value)
Mean	White tract \$513,324	African American tract \$946,661	-\$433,337 (-1.07; $p = .29$)	
Median	\$5100	\$9,962		-\$4862 (-1.16; $p = .25$)
Mean	White ZIP code \$1,106,638	African American ZIP code \$655,517	\$451,121 (1.01; $p = .31$)	
Median	\$22,125	\$11,522		\$10,603 (1.10; $p = .27$)
Mean	Non-Hispanic tract \$777,924	Hispanic tract \$1,048,900	-\$270,976 (-0.61; $p = .54$)	
Median	\$8440	\$15,000		-\$6560 (-1.38; $p = .17$)

Mean	Non-Hispanic ZIP code \$974,429	Hispanic ZIP code \$586,005	
Median	\$20,500	\$6,500	\$338,424 (0.96; $p = .34$) \$14,000 (0.23; $p = .82$)
Mean	High-income tract \$314,918	Low-income tract \$1,255,684	–\$940,766 (–2.20; $p = .03$)
Median	\$5100	\$21,750	–\$16,650 (1.35; $p = .18$)
Mean	High-Income ZIP code \$1,111,777	Low-Income ZIP code \$347,513	\$764,263 (1.96; $p = .05$)
Median	\$13,475	\$5,575	\$7900 (1.21; $p = .23$)

Note. Q1 African American tracts = 1% and Q3 African American tracts = 8%; Q1 African American ZIP codes = 1.5% and Q3 African American ZIP codes = 17.6%; Q1 Hispanic tracts = 1% and Q3 Hispanic tracts = 18%; Q1 Hispanic ZIP codes = 3% and Q3 Hispanic ZIP codes = 23%; Q1 median household income tracts = \$31,264 and Q3 median household income tracts = \$52,463; Q1 median household income ZIP codes = \$29,926 and Q3 median household income ZIP codes = \$43,696.

assessed against petroleum refineries are greater in African American compared to White tracts; Hispanic compared to non-Hispanic tracts; and, low-income compared to high-income tracts. The opposite is true for *ZIP codes*, where the mean and median penalties against petroleum refineries are greater in White compared to African American ZIP codes; non-Hispanic compared to Hispanic ZIP codes; and, high-income compared to low-income ZIP codes. These results are consistent with Atlas's (2001) finding that mean assessed penalty is greater in minority and low-income census tracts, and Lavell and Coyle's (1992) finding that the mean assessed penalty is lower in minority and low-income ZIP codes than in White and high-income ZIP codes. Also important is the fact that only median household income appears to be statistically significant ($p < .05$)—and only in the case of mean comparisons, which are likely inappropriate due to severe positive skewness of the distribution. For example, petroleum refineries in high-income tracts are assessed average penalties of \$314,918, while refineries in low-income tracts are assessed average penalties of \$1,255,684. In contrast, petroleum refineries in high-income ZIP codes are assessed average penalties of \$1,111,777, while petroleum refineries in low-income ZIP codes are assessed average penalties of \$347,513. No other statistically significant mean or median differences are evident. However, it is still necessary to adjust for important factors that may influence penalty amounts using appropriate multivariate statistical techniques.

Multivariate Findings

Initial estimation of penalty amount—a nonnegative integer—using ordinary least squares regression procedures indicated a severe problem with heteroscedasticity of the error terms. As noted, most refineries received relatively low penalties and few refineries received large penalties. Consistent with these observations and the suggestions of Long (1997), multivariate analysis was conducted using negative binomial regression. Those results are presented in Table 2.

The first model estimated assessed penalties using census tract data and relevant controls. While several variables are statistically significant in that model, *proportion African American*, *proportion Hispanic*, and *median household income* were not related to the penalty amount assessed against petroleum facilities. This result is consistent with the bivariate census-tract findings that assessed penalties are not higher in poor and minority neighborhoods.

The second model estimated assessed penalties using ZIP code data and relevant controls. In that model two variables stand out as statistically significant: *median household income* and *proportion Hispanic*. Median household income is strongly associated with assessed penalty amounts. The results suggest that each thousand-dollar increase in median household income in ZIP codes is associated with an 11.6% increase in the average penalty assessed against petroleum refineries ($p < .05$). Thus, while residents living in more affluent census tracts where environmental violations have occurred do not appear to benefit from more vigorous enforcement on the part of state agencies and the U.S. EPA, residents living in more affluent ZIP codes clearly have some advantage when other variables are held constant.

On average, the assessed penalty against refineries in Hispanic ZIP codes is nearly 95% less than the assessed penalty against refineries in ZIP codes with no Hispanic residents, suggesting that residents living in Hispanic ZIP codes do not appear to be protected via assessed penalties to the same degree as residents living in non-Hispanic ZIP codes. In contrast, we also found that residents living

TABLE 2 Percent Change in Penalty Amount Calculated from Negative Binomial Regression of Dollar Fines on Race, Ethnicity, and Median Household Income

Variable	Tract model estimate (SE)	Percent change	ZIP code model estimate (SE)	Percent change
Community demographics				
Median household income	-0.04 (0.04)	-3.9	0.11 (0.02)	11.6*
Proportion African American	-2.89 (1.98)	-94.4	-0.23 (1.67)	-20.5
Proportion Hispanic	-1.96 (1.05)	-85.9	-3.06 (1.02)	-95.31*
Enforcement history				
Significant noncompliance	0.62 (0.61)	85.9	-0.48 (0.73)	-38.1
No. previous enforcement actions	-0.10 (0.07)	-9.5	-0.21 (0.07)	-18.9*
Case characteristics				
Law violated (vs. CAA)				
CWA	-8.58 (1.56)	-100.0*	-11.53 (1.35)	-100.0*
RCRA	1.02 (0.98)	177.3	0.03 (0.87)	3.04
Multimedia case	0.66 (0.95)	93.5	0.91 (0.92)	148.4
No. facilities covered by violation	0.19 (0.16)	20.9	0.17 (0.17)	18.5
Lead agency (U.S. EPA vs. state)	3.46 (0.84)	3081.7*	3.18 (0.74)	2304.7*
Judicial outcome	0.15 (1.06)	16.2	0.36 (0.92)	43.33
Company and facility characteristics				
Annual company sales	0.01 (0.00)	1.0*	0.01 (0.03)	1.0*
No. company employees	-0.96 (0.41)	-61.7*	-1.05 (0.36)	-65.0*
Company is subsidiary	-0.69 (0.56)	-49.8	-0.28 (0.53)	-24.4
Pollution prevention ranking	-0.90 (0.93)	-59.34	-0.00 (0.83)	-0.30
Political/economic climate				
Democrat state governor	-0.21 (1.31)	-18.94	2.32 (1.16)	917.6*
Percent of gross state product	-0.80 (0.40)	-55.1*	-1.09 (0.43)	-66.4*
Region (vs. region 2)				
Region 3	0.61 (1.79)	84.0	-3.10 (1.87)	-95.4
Region 4	-2.09 (2.25)	-87.6	-3.83 (2.17)	-97.8
Region 5	-1.28 (1.89)	-72.1	-3.33 (1.76)	-96.4
Region 6	1.07 (2.18)	191.5	-2.67 (1.82)	-93.1
Region 7	-0.88 (2.88)	-58.5	-3.78 (2.90)	-97.7
Region 8	1.40 (2.17)	305.5	-2.72 (2.06)	-93.4
Region 9	-0.35 (1.49)	-29.5	-4.54 (1.66)	-98.9*
Region 10	-1.76 (1.36)	-82.8	-1.61 (1.24)	-80.0
Constant	15.9 (3.49)		9.55 (2.53)	
Log likelihood	-2145.35		-2135.24	
<i>n</i>	206		206	

Note. The percentage change is calculated using the following transformation on the independent variables $[(\exp b) - 1] * (100)$. Significant differences: * $p < 0.05$ (two-tailed).

in Hispanic census tracts may not be disadvantaged in terms of environmental enforcement.

In contrast to findings on income and ethnicity, findings for *proportion African American* are consistent across census-tract and ZIP-code models; there appears to be no relationship between proportion African American and assessed penalty.

The results in Table 2 also indicate that several additional control variables are important and consistent predictors of assessed penalty amounts. Two case characteristics predicted penalty amounts: *law violated* and *lead agency*. Penalties for violations of the Clean Air Act are much larger than penalties for violation of the Clean Water Act. This is consistent with the fact that the petroleum industry is a much more serious air polluter than water polluter. In addition, we also found that the U.S. EPA tends to assess much higher penalties on average than state agencies. In fact, the assessed penalty is nearly 2300–3000% larger when the U.S. EPA has the lead in the enforcement action.

Several company characteristics affected assessed penalties. First, *annual company sales* increased penalties by a small amount—each \$1 billion increase in sales is associated with about a 1% increase in average assessed penalty. Thus, companies with higher sales may be viewed as able to pay more for violations of environmental laws. This finding is consistent with the U.S. EPA general enforcement philosophy that any monetary gain derived from a violation should be offset by the assessed penalty.

While sales increased penalty, number of employees was associated with a decrease in assessed penalty. Each additional 1000 employees is associated with a 62–65% decrease in average assessed penalty. Thus, while larger companies have larger sales and larger penalties, that increase is more than offset by the penalty reduction associated with a larger number of employees. This finding may be the result of an informal U.S. EPA policy to assess lower fines on companies with a large number of employees. In the past, industry has successfully argued that jobs would be lost as a result of strict enforcement of environmental laws (Mintz 1995). The U.S. EPA has not been immune to this criticism.

Political/economic climate in the state also influenced penalty amounts. Assessed penalties were lower in states where the petroleum industry has a larger influence on a state's economy. Every one unit increase in *percent gross state product* is associated with a 55–66% reduction in the average penalty amount.

Two other variables were related to penalty amounts, but only in the ZIP-code model. First, assessed penalties were larger in states with Democratic governors (*Democrat state governor*), supporting the idea that party politics may influence assessed penalties. Second, previous enforcement actions (*no. previous enforcement actions*) negatively affected penalty amounts, decreasing the penalty associated with persistent violations. Reviewing the data further, we discovered that refineries with the greatest number of past enforcement actions tend to have a greater number of present enforcement actions. This finding would indicate that penalties are doing little to deter violations of law. This may be because penalties are too small to alter decision-making practices (see sales–penalty amount and employees–penalty amount findings).

In contrast to U.S. GAO conclusions, U.S. EPA region did not impact penalty assessment. This result may be the product of the additional controls that account for penalty variation, and regional variation (e.g., political economic climate,

community demographics, and enforcement history) and are included in our models.

Discussion and Conclusion

Previous research on the association between race, ethnicity, income, and penalties assessed against companies violating environmental laws is mixed. The results of our study on penalties in the petroleum industry fit a similar pattern. After controlling for enforcement history, case characteristics, company characteristics, and political/economic climate, we discovered evidence of inequity in the case of Hispanic and low-income communities—but *only if ZIP codes rather than census tracts are used to obtain the demographic data*. This finding, while consistent with previous studies of assessed penalties, is cause for question. Specifically, why is there a different result if ZIP codes rather than census tracts are studied? We can think of two potential answers to this question. First, the difference might reflect ecological fallacies and aggregation errors. Thus, the conclusions we draw at the ZIP-code level do not hold true when census tracts are studied. While this explanation is methodologically desirable it may possibly be too simplistic. For instance, it is unrealistic to assume that the potential social processes that influence enforcement practices are restricted to census-tract boundaries. As Williams (1999, 323) argues, “A census tract may be too small to measure the cultural, social, and political space that a community might occupy.” Thus, social scientists interested in larger geographic patterns of social disadvantage, such as racial and economic segregation that may easily traverse census-tract boundaries, are likely to be interested in our findings at the ZIP-code level. Because social processes tend to be rather complex, we recommend that future environmental justice research employ various units of analysis. While such multi-unit studies still leave open to debate the most appropriate unit to study, they also help to prevent sweeping assumptions about environmental justice that may be unwarranted.

Several additional findings merit comment. In the case of company characteristics, companies with more employees appear to be better at mitigating penalty amounts, while companies with larger annual sales are penalized at a higher rate. These two effects are offsetting, however, as there is some relationship between annual sales and number of employees. The political-economic climate is also an important predictor of assessed penalties, and our analysis indicated that the economic power of the petroleum industry within a given state clearly mitigates. In short, our findings concerning company characteristics and the political-economic climate broaden the kinds of questions that researchers should investigate when examining penalties assessed for violation of environmental laws (Simon 2000).

Our findings also relate to the effects of fines as deterrents, and the idea that repeat violators are treated more severely. Contrary to common assumptions, we found that the number of previous enforcement actions reduced penalties. It may be that past enforcement actions have altered the behavior of these companies and present violations are less serious as a result. Further research is needed to examine whether this is, in fact, occurring.

There are two major drawbacks of this research. First, we are unable to put forth any broad conclusions regarding the existence of environmental justice. We have only studied one industry. It could be the case, for instance, that there is substantial industry variation in patterns of environmental justice. Moreover,

we have only studied variation in monetary penalties assessed against petroleum refineries. Thus, our findings cannot be generalized to all available enforcement mechanisms.

Second, like DOCKET, ECHO suffers from measurement error. For instance, representatives from the American Petroleum Institute have identified several errors in ECHO (Retzsch 2003). The most serious of these errors is that there are some reports of enforcement actions at facilities where no enforcement actions occurred (Retzsch 2003, 4). However, the report by the American Petroleum Institute suggests that ECHO error is random. If the measurement error in ECHO is random, then the results presented here are likely too conservative. That is, random error in our dependent variable is likely to weaken the relationship between race, ethnicity, income, and assessed penalties.

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